

CURRENT GREEN HYDROGEN ECOSYSTEM

65-70
MW
Operational capacity for electrolyser

21
NGHM-incentivised projects with 3 GW capacity for electrolysers

19
NGHM-incentivised green hydrogen projects

12,000
TONNES
Operational capacity for green hydrogen

Over 80
Pilot projects, of which over 44 are for green hydrogen and derivatives. These include a green hydrogen train, Adani Group's five MW hydrogen plant, and Maruti Suzuki's **300 KW** electrolyser plant

SOURCES MNRE, GH2 INDIA, TRADE NEWS



"Unless India aligns with the EU regulatory requirements for green hydrogen, it would be hard to export to regions like Europe"

GOPAL SARANGI
Associate Professor, TERI School of Advanced Studies

its derivatives is expected to be over 100 MT by 2030 and India is "aiming at" capturing about 10% of it (or 10 MT).

But the global market has a more complicated story to tell.

In July and August 2026, media reports said that investors were cutting back on investment and capacity creation plans for both electrolysers and green hydrogen production. In 2025, media reported that investors were scaling back and scrapping projects across Europe, the US, Australia, and Japan due to high costs, weak demand, and other factors.

With domestic demand yet to take off, India is looking to export green hydrogen. But exporting has its own challenges. Energy expert Gopal Sarangi of the TERI School of Advanced Studies points to three stringent conditions the EU has put for importing green hydrogen, which are also valid for its domestic producers: "new" renewable energy (electricity) generation, hourly matching of production and consumption, and geographic correlation between the renewable energy source powering the electrolyser and hydrogen production.

In contrast, India's 2025 certification scheme defines green hydrogen only in terms of carbon emissions—not more than 2 kg carbon dioxide equivalent per kg of hydrogen.

"Unless India aligns with the EU regulatory requirements for green hydrogen or develops some strategic approach, such as a coalition of like-minded countries in developing a common green hydrogen taxonomy and regulatory framework, it would be hard to export to regions like Europe," says Sarangi.

An IEEFA-JMK research report in September pointed out that solar PV plants are operating at 35-40% capacity—well below the 50-65% level required for sustainable operations.

India's solar PV experience shows the risks of building manufacturing capacity faster than demand and the supporting supply chain can keep pace. The green-hydrogen push may therefore require a different sequencing, with demand creation keeping pace with manufacturing capacity. BT